Assignment 3: Exploratory Data Analysis & Descriptor/Fingerprint Calculation
Intern Name: Abdulrahman Olamilekan Raji
Summary Report on Exploratory Data Analysis, Descriptor Statistics, and Fingerprint Calculation

The work aimed to explore the chemical and bioactivity characteristics of curated compounds and prepare a high-quality dataset suitable for QSAR modelling. The workflow involved data cleaning, exploratory data analysis (EDA), calculation of Lipinski descriptors, statistical evaluation, and generation of molecular fingerprints for the previously RAD51 protein dataset from assignment 2 (ChEMBL Target ID: CHEMBL2034807).

Following preprocessing, the dataset was reduced from 130 initial records to 70 unique compounds after removal of missing values, intermediate activity entries, and duplicate SMILES aggregation. IC50 values were converted to pIC50 to normalize bioactivity data and allow better statistical interpretation. Compounds were classified into active and inactive groups using a pIC50 threshold of 6.

Exploratory data analysis revealed several key findings. The histogram of pIC50 values was shown. The bar chart of bioactivity classes demonstrated a biased dataset, which can affect predictive models (with less than 10 active compounds and over 60 inactive compounds). Boxplot visualization showed a clear separation between active and inactive compounds based on pIC50, confirming the appropriateness of the classification threshold. Moreover, the chemical space scatter plot of molecular weight versus LogP was also indicated.

Lipinski descriptors were calculated to evaluate drug-likeness based on physicochemical properties. Statistical analysis showed that most compounds satisfied Lipinski's Rule of Five, with molecular weights generally below 500 Da, LogP values below 5, and acceptable hydrogen bond donor and acceptor counts. Mann-Whitney U tests indicated that pIC50 displayed a statistically significant difference between active and inactive groups. Among physicochemical descriptors, hydrogen bond donors showed moderate statistical significance, while molecular weight, LogP, and hydrogen bond acceptors demonstrated weaker discriminatory power between bioactivity classes.

For molecular fingerprint generation, the PaDELPy tool was used. This software is a Python wrapper for the PaDEL-Descriptor program and allows automated calculation of large-scale molecular descriptors and fingerprints. Using this method, 881 PubChem fingerprint bits were generated for each compound, representing binary structural features that encode the presence or absence of specific chemical substructures.

This method was selected because it provides comprehensive descriptor coverage, is widely used in QSAR studies, and allows reproducible, high-throughput calculation of fingerprints directly from SMILES input. Additionally, PaDELPy integrates easily with Python workflows and supports multiple descriptor types, making it suitable for machine learning-based drug discovery analysis.